

Day 19 - Lecture 19.7: Subplots in Matplotlib

1. Overview & Purpose

When communicating findings or building dashboards, putting all curves on a single plot can cause severe visual clutter and scale mismatch. **Subplots** allow us to place multiple separate graphs in a clean, organized **grid layout** within a single figure.

2. Syntax & Row-Major Indexing

```
plt.subplot(nrows, ncols, index)
```

- `nrows` : Total number of rows in the grid.
- `ncols` : Total number of columns in the grid.
- `index` : 1-based index specifying which subplot is currently active.

2x2 Subplot Grid (Row-Major Order)

`plt.subplot(2, 2, 1)`
Top-Left

`plt.subplot(2, 2, 2)`
Top-Right

`plt.subplot(2, 2, 3)`
Bottom-Left

`plt.subplot(2, 2, 4)`
Bottom-Right

[!IMPORTANT] `index` starts at 1 (not 0) and counts row-wise from top-left to bottom-right. Shortcut notation: `plt.subplot(221)` is identical to `plt.subplot(2, 2, 1)`.

3. The Role of `plt.tight_layout()`

When creating multiple subplots, axis labels, tick marks, and titles from adjacent subplots often collide or overlap. Calling `plt.tight_layout()` automatically recalculates subplot margins, paddings, and font metrics to ensure zero overlap.

4. Code Implementation: Function Transformations

```
import matplotlib.pyplot as plt
import numpy as np

x = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
y1 = [np.sqrt(i) for i in x] # Square root
y2 = [i * 2 for i in x]      # Linear / Double
y3 = [i ** 2 for i in x]     # Quadratic / Squares
y4 = [i ** 3 for i in x]     # Cubic / Cubes

plt.figure(figsize=(10, 8))

# Subplot 1: Square Root
plt.subplot(2, 2, 1)
plt.plot(x, y1, 'b-o')
plt.title("Plot 1: Square Root ( $\sqrt{x}$ )", fontweight="bold")
plt.grid(True)

# Subplot 2: Linear
plt.subplot(2, 2, 2)
plt.plot(x, y2, 'g-s')
plt.title("Plot 2: Linear ( $2x$ )", fontweight="bold")
plt.grid(True)

# Subplot 3: Quadratic
plt.subplot(2, 2, 3)
plt.plot(x, y3, 'r-^')
plt.title("Plot 3: Quadratic ( $x^2$ )", fontweight="bold")
plt.grid(True)

# Subplot 4: Cubic
plt.subplot(2, 2, 4)
plt.plot(x, y4, 'm-d')
plt.title("Plot 4: Cubic ( $x^3$ )", fontweight="bold")
plt.grid(True)

plt.tight_layout()
plt.show()
```

5. Limitations of Stateful `plt.subplot()`

The stateful approach relies on a global pointer to the "current active axis". For complex multi-panel figures or interactive code, this quickly leads to bugs. In the

next lecture, we learn the superior **Object-Oriented API** (`fig, axes = plt.subplots()`).